

KNATWARE TECHNOLOGY PRESENTS

THE FINE-TUNING CYCLE!

Starring KNAT the fine-tuning gnat, based scene-for-scene on `Knatware_LLM_FineTuning_V2_Colab.ipynb` — every panel maps to a real section of the notebook.



1

SETUP

VROOM?!

"Runtime > Change runtime type > GPU... please tell me I got a GPU!"

Section 1 — Environment Setup: runs `!nvidia-smi` to confirm a GPU is attached, then installs `transformers`, `datasets`, `accelerate`, `peft`, `trl` & `bitsandbytes`.



2

SETUP

BEEP BOOP

"Hugging Face badge? Check. W&B badge? Optional, but I like dashboards."

Section 2 — HF & W&B Login: optional `login()` for gated models (Llama, Gemma) or pushing to the Hub, plus optional `Weights & Biases` tracking.



3

SETUP

CLICK. CLICK.

"Model: Qwen2.5-1.5B. Dataset: guanaco-llama2-1k. Everything downstream reads from ME."

Section 3 — Core Config: sets `MODEL_NAME`, `DATASET_NAME`, `OUTPUT_DIR`, `MAX_SEQ_LENGTH=1024` and `USE_4BIT=True` in one control block.



4

DATA

SHOVEL SHOVEL

"Load the dataset, mash instruction + response into one tidy 'text' column!"

Section 4 — Load & Preprocess: pulls the dataset from the Hub (or a local file) and, if needed, formats fields into the single `text` field `SFTTrainer` expects.



5

MODEL

SQUEEEEEEZE!

"Big model, tiny GPU. Time to shrink it to 4-bit — QLoRA style!"

Section 5 — Tokenizer & Base Model: loads the tokenizer, sets a pad token, and loads the base model in 4-bit precision (BitsAndBytesConfig) to fit a free-tier T4.



6

TUNE

STICK! STICK!

"Freeze the big guy solid — I'll just bolt on some tiny trainable patches."

Section 6 — Configure LoRA: freezes base weights, attaches low-rank adapters ($r=16$, $\alpha=32$, $\text{dropout}=0.05$) onto the q/k/v/o projection layers. Only ~1-2% of params train.



7

TUNE

TWEAK TWEAK

"Learning rate: $2e-4$. Batch: 2, x4 grad accumulation. Cosine schedule. Chef's kiss."

Section 7 — Training Args: sets epochs=3, batch size, gradient accumulation, learning rate, warmup, cosine LR schedule, bf16, and gradient checkpointing — the real "tuning" knobs.



8

TRAIN

HUFF! PUFF! PEDAL!

"trainer.train()... this is the long one. Loss, go DOWN!"

Section 8 — Build Trainer & Train: SFTTrainer wraps model + args + dataset + LoRA config, then `trainer.train()` runs the actual fine-tuning loop, logging loss every 10 steps.

9

SHIP

SEALED & SAVED!

"Adapter's tiny — a few MB! Save it, maybe merge it, maybe mail it to the Hub."

Sections 9-10 — Save & Push: saves the small LoRA adapter + tokenizer to `OUTPUT_DIR`; optional merge into a standalone model; optional `push_to_hub()`.

10

SHIP

DOES IT WORK?!

"Load base model + adapter, ask it a question... ooh, that's actually good!"

Section 11 — Quick Inference Test: reloads the base model, attaches the fine-tuned adapter, and generates a sample response as a sanity check.

11

TUNE

SPIN AGAIN!

"Not happy with the loss? Sweep a few learning rates and LoRA ranks, compare, repeat!"

Section 12 — Hyperparameter Sweep (optional): a small grid-search loop over settings like learning rate to compare training loss across mini-runs.

THE CYCLE CONTINUES!

Not good enough yet? Knат hops back to Panel 3 or 7, tweaks the config or hyperparameters, and runs the whole loop again — that's the "tuning" in fine-tuning.

REMEMBER IT WITH: F.I.N.E. T.U.N.E.

The 8 core panels spell out the very word this notebook is all about. Say it out loud before you open Colab.

F

FIRE UP

Panel 1 — check GPU, install libraries.

I

IDENTIFY

Panel 2 — log in to Hugging Face / W&B.

N

NAME THE CONFIG

Panel 3 — model, dataset, output dir, seq length.

E

EXTRACT DATA

Panel 4 — load + format the dataset's "text" field.

T

TRANSFORM IN 4-BIT

Panel 5 — load tokenizer + QLoRA base model.

U

UPGRADE W/ LORA

Panel 6 — freeze base, attach adapter patches.

N

NAIL THE KNOBS

Panel 7 — learning rate, epochs, batch, schedule.

E

ENGAGE!

Panel 8 — `trainer.train()` actually runs.

Then comes **S.T.S.** — **S**ave, **T**est, **S**weep (Panels 9-11) — and if the sweep says "try again," you loop back to **Name-the-config** or **Nail-the-knobs** and re-run the cycle.

MEET THE CAST & CREW

KNAT

The mascot. A gnat-shaped robot who represents the whole notebook — small, fast, and surprisingly good at squeezing into tight GPU memory.

THE BIG FROZEN MODEL

Qwen2.5-1.5B-Instruct, loaded in 4-bit. Doesn't move (frozen weights) — Knat's LoRA patches do all the learning.

THE ADAPTER PATCHES

The small LoRA matrices ($r=16$) that get trained. Tiny in file size, big in effect.

THE DATA HOPPER

The guanaco-llama2-1k dataset, formatted into one clean "text" field before training.

THE TREADMILL

SFTTrainer — where 3 epochs of actual training happen, logging loss the whole way.

THE HUB

Hugging Face Hub — optional final destination for the finished adapter.

Comic guide to Knatware_LLM_FineTuning_V2_Colab.ipynb · kayodekosi/Knat-LLM · Made for learning, not for training — go run the real notebook after this!